TD3 — Working as a team on GitHub

Git, Python Environments & Coding Best Practices

MSA-DATB07-03 — 1 h 30 practical — **answer key**

Learning outcome. Clone a team repository, split the work with Issues, push a branch, open a pull request, review a teammate's pull request and merge it, resolve a conflict in a pull request, and see branch protection at work (a pull request cannot be merged without an approval).

Tip — how this TD works

You work in teams of 3 or 4, in one GitHub repository per team: a pull request needs a teammate to review it. As in TD2: **guided** steps give the commands, **Your turn** steps ask you to adapt them, and ★ marks the key questions, the kind of reasoning asked in the quiz. Everything of today goes into a folder `td3`, in your course folder (next to `td2`). Files are created and edited in VS Code (method 1 of TD2); the terminal commands of method 2 work too. This TD is not graded.

Tip — reminder from TD2: the loop, and branches

Today everything goes through these commands, now with teammates:

- **save a change**: edit and save the file (VS Code) → `git status` (what changed?) → `git add <file>` (put it in the next commit) → `git commit -m "message"` (save it in the history) → `git log --oneline`
- **a branch**: `git switch -c <name>` (create it and go to it), `git switch main` (back on `main`), `git merge <branch>` (bring a branch into the one you are on)
- **a conflict**: keep the right lines, delete the three marker lines, save → `git add` → `git commit`

New today: `git push` sends your commits to GitHub, `git pull` brings back those of your teammates.

Part A — Your team and its repository *(20 min)*

Exercise 1 — Form the team, create the repository

Form teams of 3 or 4, with the people sitting near you. One member is the **owner**: they create the repository, the others are **collaborators**. Everybody needs the GitHub account of TD2, signed in on https://github.com.

1. **Owner only.** Open https://github.com/msa-datb07-git/team-project-template, click *Use this template → Create a new repository*. Owner: your account; name: `msa-datb07-team`; visibility: **Public**.
2. **Owner only.** *Settings → Collaborators → Add people*: add each teammate with their GitHub user name. **Teammates**: accept the invitation (e-mail, or the bell icon on GitHub).
3. **Owner only.** Protect `main`: *Settings → Branches → Add classic branch protection rule*. Branch name pattern `main`; tick *Require a pull request before merging* and *Require approvals* (1); at the bottom, tick *Do not allow bypassing the above settings* (otherwise the owner could still push on `main`). Save.

Exercise 2 — Clone it

First, a folder for TD3. Go to your course folder, the one of TD2 (the folder that contains `td2`), and create `td3` in it:

```
cd ~/Documents/courses/git        # YOUR course folder: adapt the path (or "cd ", drag
    the folder, Enter)
pwd                               # check: you are in your course folder
mkdir td3
```

```
cd td3
```

Tip — your course folder is elsewhere, or you have none

- **Elsewhere** (Desktop, OneDrive…): type `cd` and a space, drag that folder from the file explorer onto the terminal, press `Enter`. Then `mkdir td3` and `cd td3`.
- **None**: create one, with `td3` inside, in one command: `mkdir -p ~/Documents/courses/git/td3` (`-p`: the whole chain of folders), then `cd ~/Documents/courses/git/td3`.
- In the "Where you should be" boxes below, replace `~/Documents/courses/git` by your own course folder.

Guided, everybody. On the team repository, the green *Code* button gives the address (*HTTPS* tab). In `td3`, predict, run, explain:

```
git clone https://github.com/<owner>/msa-datb07-team.git
cd msa-datb07-team
git remote -v                              # (1)
git log --oneline                          # (2)
python -m pip install pytest               # macOS: python3 -m pip
python -m pytest                           # (3)
```

1. In (1), what is `origin`? In (2), how many commits does the repository have?
2. In (3), how many tests pass?
3. Open the folder `msa-datb07-team` in VS Code (*File* → *Open Folder*) and read `TASKS.md`: what is the task of each member of your team today?

Answer

`origin` is the name of the GitHub repository the clone came from: `git remote -v` prints its address twice (fetch and push). 4 commits (*Use this template* copies the files, with a fresh history), 3 tests pass. `TASKS.md`: T1 to T4, one per member, each with the files to change; the code is in `corrections/`.

Instructor note

Check that the repository is *public* (branch protection is free only on public repositories) and that every member accepted the invitation before Part C: otherwise their push is refused (403, "Permission denied").

Part B — Split the work with Issues *(10 min)*

Exercise 3 — One Issue per task

Where you should be: `pwd` ends with `courses/git/td3/msa-datb07-team`
If not: `cd ~/Documents/courses/git/td3/msa-datb07-team`

An **Issue** is a card in the to-do list of the repository: everyone sees what must be done, and who does it.

1. Share tasks T1 to T4 of `TASKS.md`: one per member. T1 and T2 go to two different people, T3 and T4 too. Teams of 3: one member also takes T4, after their first task.
2. **Each member**, on GitHub: tab *Issues* → *New issue*. Title: the title of your task (e.g. `T1 --- add average_order_value`). On the right, *Assignees* → *assign yourself*. *Create*. Note the number GitHub gives it: `#1`, `#2`…

Your turn Open the *Issues* tab: who does what in your team? Which Issue number is yours?

Part C — One task each, through pull requests *(55 min)*

Exercise 4 — Branch, commit, push, pull request

Where you should be: `pwd` ends with `courses/git/td3/msa-datb07-team`
If not: `cd ~/Documents/courses/git/td3/msa-datb07-team`

Guided — the author. Example with T1; use the branch name and the message of your task:

```
git switch main
git pull                                        # start from the latest main
git switch -c feat/average-order-value          # a branch for your task
```

Do the task in VS Code, following `TASKS.md` step by step (code, test, README), save. Stuck? The folder `corrections/` gives the code of every task: try first, then compare or copy. Then:

```
python -m pytest                                # the tests must pass
git add -A                                      # -A = all the changes
git commit -m "feat(metrics): add average_order_value"
git push -u origin feat/average-order-value     # -u: remember where this branch goes
```

Pitfall — the first push asks you to sign in to GitHub

- **Windows**: a window opens, *Sign in with your browser*: sign in to GitHub and authorise. Git remembers it.
- **macOS**: the terminal asks `Username`: your GitHub user name. `Password`: **not** your GitHub password, a *token*. On GitHub: avatar → *Settings* → *Developer settings* → *Personal access tokens* → *Tokens (classic)* → *Generate new token (classic)*, tick *repo*, generate, copy it (shown only once), paste it as the password. Nothing shows while you paste: normal, press `Enter`. The Keychain remembers it. A token is a password: never write it in a file.
- **"Permission denied" or 403**: you are signed in but not a collaborator: accept the invitation (Part A).

Git prints a link: open it, or click *Compare & pull request* on GitHub. Fill in the template: *What*, *Why* with `Closes #1` (your Issue number: merging the pull request will close the Issue). Choose a teammate in *Reviewers*. *Create pull request*.

Guided — the reviewer. Open the pull request, tab *Files changed*. Click on a line to comment: a question, a suggestion, a possible bug. Then *Review changes* → *Approve*, or *Request changes* if something must change first (the author fixes it with a new commit on the same branch and pushes again: the pull request updates itself).

Guided — after the approval. The author clicks *Merge pull request*, then *Delete branch*. Everybody brings `main` up to date:

```
git switch main
git pull
```

Pitfall — git pull: "Need to specify how to reconcile divergent branches"

Mostly on macOS, when you have a commit on `main` that GitHub does not have (a commit made on `main` by mistake). Run `git pull --no-rebase`: download and merge, this time only, nothing is changed in your settings. (To make it the default: `git config --global pull.rebase false`; to undo that: `git config --global --unset pull.rebase`.)

Answer

"Merging is blocked" on the pull request: the approval is missing (the protection works). A push refused with "Permission denied" or 403: the invitation to the repository was not accepted. What makes a good pull request: a clear message, the template filled in, a reviewer chosen, at least one real comment from the reviewer, merged after approval.

Answer

Reference code of the four tasks (the full end state is in `solutions/team-project/`; after T1 to T4, `python -m pytest` prints `5 passed`).
T1 — in `src/sales_analysis/metrics.py`, at the end:

```
def average_order_value(rows: list[dict]) -> float:
    """Total revenue divided by the number of orders (one row = one order)."""
    if not rows:
        return 0.0
    return round(total_revenue(rows) / len(rows), 2)
```

In `__init__.py`: add `average_order_value` to the `from sales_analysis.metrics import ...` line and `"average_order_value"` to `__all__`. In `tests/test_metrics.py`:

```
def test_average_order_value():
    from sales_analysis import average_order_value

    assert average_order_value(ROWS) == 26.83      # 80.5 / 3 orders
    assert average_order_value([]) == 0.0
```

README, list *Features*: `- average order value`.
T2 — in `metrics.py` (`defaultdict` is already imported at the top of the file):

```
def top_products(rows: list[dict], n: int = 3) -> list[str]:
    """The n products with the highest revenue, best first."""
    totals: dict[str, float] = defaultdict(float)
    for r in rows:
        totals[r["product"]] += r["quantity"] * r["unit_price"]
    return [p for p, _ in sorted(totals.items(), key=lambda item: item[1], reverse=
    True)[:n]]
```

Same changes in `__init__.py` with `top_products`; README: `- top products`; test:

```
def test_top_products():
    from sales_analysis import top_products

    assert top_products(ROWS, n=2) == ["Gizmo", "Gadget"]   # 35.0, then 25.5
```

T3 — run once, then README at the end:

```
PYTHONPATH=src python -c "from sales_analysis import load_sales, total_revenue;
    print(total_revenue(load_sales('data/sales.csv')))"
# 6945.5
```

```
## Usage

Install pytest and run the tests:
    python -m pip install pytest
    python -m pytest

Compute the total revenue of the sales file:
    PYTHONPATH=src python -c "from sales_analysis import ..."   (the command above)
```

T4 — README, at the end:

```
## Data dictionary

| column     | type               | example    |
|------------|--------------------|------------|
| date       | date (YYYY-MM-DD)  | 2024-01-09 |
| product    | text               | Gadget     |
| quantity   | integer            | 4          |
| unit_price | decimal (EUR)      | 24.00      |
| region     | text               | West       |
```

and the *Team* section: the names and GitHub user names of the members.

Exercise 5 — The planned conflicts

Where you should be: `pwd` ends with `courses/git/td3/msa-datb07-team`
If not: `cd ~/Documents/courses/git/td3/msa-datb07-team`

T1 and T2 change the same lines (`__init__.py`, the *Features* list of the README); T3 and T4 both add a section at the end of the README. Once the first pull request of a pair is merged, the second one shows *This branch has conflicts that must be resolved.*
Your turn ★ The author of the second pull request resolves the conflict *on their branch*, with what you did in TD2: bring the new `main` into the branch, resolve, finish the merge, push. Keep *both* changes (both functions, both bullets, both sections), run the tests.

Answer

```
git switch main
git pull                              # get the merged pull request of your teammate
git switch feat/top-products
git merge main                        # CONFLICT in __init__.py, metrics.py,
    test_metrics.py, README.md
# in VS Code: keep both sides of each block, delete the marker lines, save
python -m pytest                      # must pass
git add -A
git commit                            # the merge commit: save, close the tab
git push                              # the pull request updates itself: no conflict
    any more
```

In `__init__.py`, a correct resolution imports both functions and lists both in `__all__`. Also correct: GitHub's *Resolve conflicts* button, which creates the same merge commit. Wrong: `git push --force`, or markers left in a file (the tests fail).

Exercise 6 — Read the team history

Where you should be: `pwd` ends with `courses/git/td3/msa-datb07-team`
If not: `cd ~/Documents/courses/git/td3/msa-datb07-team`

Your turn ★ Once the four pull requests are merged: `git switch main`, `git pull`, `git lg`. Find on the graph: one merge commit per pull request (message `Merge pull request #...`), and the merge commit where a conflict was resolved (message `Merge branch 'main' into ...`). Why does `git lg` show labels starting with `origin/`?

Answer

Each pull request merged with the *Merge* button creates a commit `Merge pull request #n from <owner>/<branch>`. The conflict resolutions are the commits `Merge branch 'main' into feat/top-products` (and the one of T3/T4), made on the branches before their pull requests were merged. `origin/main`, `origin/feat/...`: the branches as they are on GitHub, as last seen by `git pull` (deleted branches disappear after `git fetch --prune`).

Part D — Look at your work on GitHub *(5 min)*

1. On your team repository: *Pull requests* → *Closed*, *Issues* → *Closed*, and *Insights* → *Contributors*. Who did what, who reviewed what: GitHub keeps all of it, with names and dates.

2. **Before TD4**: the four pull requests (T1 to T4) should be merged into `main`, each one after a teammate approved it. A pull request still open at the end of the TD? Finish it at the start of TD4: TD4 continues in the same repository, with the same team.

Tip — to go further

- **Learn Git Branching**: `https://learngitbranching.js.org`. Git in your browser, nothing to install: you type the commands and the commit graph draws itself. Start with *Introduction Sequence*, then *Ramping Up*. Available in French (menu at the bottom left).
- **shell-practice**: `https://github.com/msa-datb07-git/shell-practice`. 33 short exercises on the terminal and Git, with their solutions: `git clone` it into your course folder.

Deliverable

Nothing to send: TD3 is not graded. The ★ questions are the kind of reasoning asked in the quiz. If you want to revise, note your answers in a file `td3_answers.md` in your `td3` folder, *next to* the repository, not inside it: a file inside the repository would end up in your next commit.